Asgnt 1:

Grade: Check Plus

x	$f(x) = P(X = x)$
10	0.10
20	0.15
40	0.30
60	0.15
80	0.25
100	0.05

At least 60 Means

(a) $P(X \geq 60) = P(60) + P(80) + P(100) = 0.15 + 0.25 + 0.05 = \underline{0.45}$ ✓Qa

(b) $E(X) = 50$

$SD = 26.27$

1. Expected Value (Mean):

$$\boxed{\mu = E(X) = \sum X f(x)}$$

$10\ (0.10) = 1$
$20\ (0.15) = 3$
$40\ (0.30) = 12$
$60\ (0.15) = 9$
$80\ (0.25) = 20$
$100\ (0.05) = 5$

$\mu = 1 + 3 + 12 + 9 + 20 + 5 = \underline{50}$ ✓ Qb

2. SD:

$$\boxed{\text{Variance}(X) = E(X^2) - \mu^2}$$

$$\boxed{SD = \sigma = \sqrt{Var(X)}}$$

$E(X^2) = \sum X^2 f(x)$:

$10^2\ (0.10) = 10$
$20^2\ (0.15) = 60$
$40^2\ (0.30) = 480$
$60^2\ (0.15) = 540$
$80^2\ (0.25) = 1600$
$100^2\ (0.0\ \ ) = 500$

$E(X^2) = 10 + 60 + \ldots + 500 = 3{,}190$

$Var(X) = 3{,}190 - 50^2 = 3190 - 2500 = 690$

$SD = \sqrt{690} = \underline{26.27}$

(c) Mean ≠ Actual Value

\$50 is a weighted average (expected value). Even though no one can withdraw exactly \$50, if thousands of customers use ATM w/ these Probabilities, the average withdrawal will approach \$50. ✓ Qc

Expected value is the long-run average, not a possible one single withdrawal.

(d) Prob within 1 SD of the Mean:

$$\boxed{\mu - SD \leq X \leq \mu + SD}$$

$50 - 26.27 \leq X \leq 50 + 26.27$ ✓ Qd

$23.73 \leq X \leq 76.27$

10
20
40
60
80
100

$P(40) + P(60) = 0.30 + 0.15 = \underline{0.45}$

(e) Prob within 2 SD of the Mean: ✓ Qe

$$\boxed{\mu - 2SD \leq X \leq \mu + 2SD}$$

$50 - 2(26.27) \leq X \leq 50 + 2(26.27)$

$-2.54 \leq X \leq 102.54$

All possible ATM withdrawals are within this range, so Prob. is everything : 1.00.